

Predictive-Maintenance Air Purifier

Innovation dossier - 1. The idea

PROPOSITION

A concept for the real 'Reliability / stopped working' friction, built with the PREDICT design move - reactive -> anticipatory.

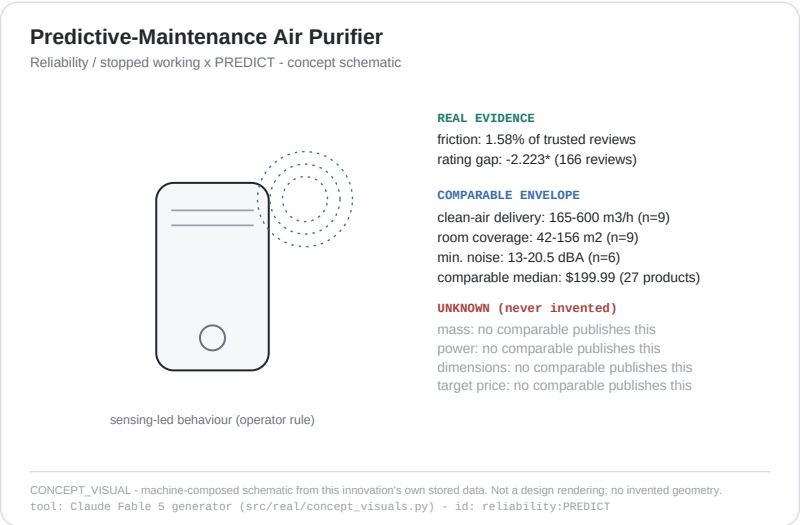


Figure: **CONCEPT_VISUAL**

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

artifact visual-reliability:PREDICT - /concept-visuals/reliability_PREDICT.svg

ENGINEERING ENVELOPE

comparable basis: 10 verified official Versuni/Philips air-purifier families (room-scale)

metric	value from verified comparables	epistemic type
performance cadr m3h	165-600 m3/h (n=9 comparables)	OBSERVED_COMPARABLE
room coverage m2	42-156 m2 (n=9 comparables)	OBSERVED_COMPARABLE
acoustic min dba	13-20.5 dBA (n=6 comparables)	OBSERVED_COMPARABLE
target mass kg	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
target power w	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
target dimensions	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
reference market price usd	median \$199.99 (range \$16.99-\$855.0, n=27) - market price of affected products today, not a proposed price	REFERENCE_MARKET_PRICE

State: **CHALLENGED**

Critic verdict CHALLENGE - a real dimension pushes back.

WHY NOW - THE OBSERVED REALITY

1.58% of trusted reviews (166 reviews across 60 real products) carry this friction, with a -2.223* average rating gap. Signal: 'Stopped working' or failing early is the most severe complaint theme by average rating gap.

TARGET USER / CONTEXT

Households whose real reviews carry this friction: 166 reviews across 60 real products (85.5% verified purchase, 2006-04-20-2023-08-27).

UNKNOWN - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

MECHANISM - METHOD CHOICE

PREDICT - Reactive -> anticipatory.

METHOD_CHOICE - authored design vocabulary

PRODUCT ARCHETYPE - DESIGN RULE

connected: Likely - operator PREDICT (reactive -> anticipatory) requires sensing/data.

room scale, mobility, size class, power class, noise ambition, maintenance burden, filter service dependency: UNKNOWN - no real data in this pipeline for this field.

DESIGN_RULE - each populated field follows from the operator's own fixed definition, never from evidence

THE DERIVATION - THREE PARTS, EACH LABELLED

1. Reality **OBSERVED**

1.58% of trusted reviews (166 reviews across 60 real products) carry this friction, with a -2.223* average rating gap. Signal: 'Stopped working' or failing early is the most severe complaint theme by average rating gap.

2. Transformation **METHOD_CHOICE**

METHOD CHOICE, not evidence: the authored operator table assigns PREDICT - "Reactive -> anticipatory." - to this friction. Analyst design judgment, applied deterministically.

3. Product consequence **FEASIBILITY_PRECEDENT**

Feasibility precedent (from the trend corpus, not a consumer-evidence consequence): Two real, already-standardized frameworks exist to build on directly: TC-R02 (ENERGY STAR already requires measured, verified ongoing performance criteria - a real regulatory template for an ongoing-performance-verified claim) and TC-R08 (Matter's Air Quality Sensor cluster is an already-standardized connectivity layer that could carry a self-diagnostic/runtime health signal without inventing a new protocol).

PARENT PATHS

No parent paths - this innovation cites no tension or assumption paths; its grounding is the friction evidence on page 1.

evidence ids: taxonomy:reliability, TC-R02, TC-R08

DESIGN DNA

F consumer friction (taxonomy:reliability) **PRESENT**

Real Amazon review-text friction theme (see Consumer Pain methodology).

O design operator **METHOD_CHOICE**

Authored operator vocabulary applied deterministically (METHOD_CHOICE, not evidence) - see OPERATORS/THEME_OPERATORS in src/real/magic_box_real.py.

S signal (reliability) **PRESENT**

'Stopped working' or failing early is the most severe complaint theme by average rating gap.

T scientific tension **MISSING_UNVERIFIED**

No real research tension shares a paper with this theme's signal evidence.

R rival gap **PRESENT**

7 named real competitors measurably weaker on this theme. [Envion, EdenPURE, Himalayan, Egofine, Belita Amy, MOONQUAKE, ZOCALOK]

C versuni capability **MISSING_UNVERIFIED**

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

A category assumption **MISSING_UNVERIFIED**

No category assumption shares a paper with this theme's signal evidence.

E economic condition **PRESENT**

Real price-weighted exposure computed for this theme (\$26,357.86).

ECONOMICS - RELATIVE INDICATORS ONLY

price-weighted exposure: \$26357.86

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

comparable market median: \$199.99 (n=27 products) - not this concept's price

MEDIAN real listed price across the 27 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

category assumptions: No category assumption shares a paper with this theme's signal evidence.

UNCERTAINTIES

- No real Versuni R&D-capacity or hardware-warranty-claims data - the rating reflects external precedent, not an internal build-cost estimate.

CONTRADICTIONS

- none recorded for this innovation

OBSERVED VS ESTIMATED VS UNKNOWN

observed: friction stats; envelope: performance cadr m3h, room coverage m2, acoustic min dba; DNA F/S/R/E

method choice: operator, archetype design rule; DNA O

unknown: envelope: target mass kg, target power w, target dimensions; persona; DNA T/C/A

Where this idea stands: **CHALLENGED**

Critic verdict CHALLENGE - a real dimension pushes back.

PROTOTYPE STATE

CONCEPT_VISUAL

CONCEPT_VISUAL means a schematic exists, nothing more: no digital model, no physical prototype, no test data.

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

NEXT EXPERIMENT

A real internal engineering estimate showing the self-diagnostic sensor suite costs materially more than a typical product refresh would lower this rating.

KILL CRITERION

No authored kill criterion yet - a non-formal innovation earns one by displacing the formal recommendation (Scenario lens) or by a human decision to invest.

CRITIC DIMENSIONS

EVIDENCE	CHALLENGE	Gate passed but prevalence (1.58%) is only marginally above the materiality floor - thin evidence base.
ECONOMIC	SURVIVE	Real price-weighted exposure (\$26,358) is at or above the corpus median (\$16,113).
COMPETITIVE	SURVIVE	Real white space: 7 named real rivals are measurably weaker on this theme.
HUMAN	NEEDS_EVIDENCE	No real usability/human-factors evidence exists in this pipeline for this concept.
PHYSICAL	NEEDS_EVIDENCE	No real lab/engineering feasibility data exists in this pipeline for this concept.
VERSUNI_FIT	NEEDS_EVIDENCE	No real organizational-capability assessment exists in this pipeline for this concept.
TIMING	NEEDS_EVIDENCE	No real market-timing/window evidence exists in this pipeline for this concept.
ROBUSTNESS	NEEDS_EVIDENCE	No real sensitivity/robustness study exists in this pipeline for this concept.

critic overall: CHALLENGE

CRITERIA RESULTS

NEEDS_EVIDENCE: 36 - PASS: 11 - N/A: 4 - CHALLENGE: 1

E3 CHALLENGE Signal state SINGLE_SOURCE_FAMILY - real, but not yet independently corroborated.

LINEAGE / PROVENANCE

generated by: src/real/innovations_real.py

magic run input sha256: 0b3f9862e0e270a295eacd709182802e981090e817f4f1072000e37aa83f150c

funnel snapshot: 73aa128628a9304d744c80cb718ecf82ed99b270b6e5d457a0198b59b6041e37

figure source: /concept-visuals/reliability_PREDICT.svg (generated 2026-08-31T16:09:19.518923+00:00 by Claude Fable 5 generator - src/real/concept_visuals.py)

data source: data/processed/innovations_real.json + data/processed/funnel_real.json (deterministic; no invented fields)

dossier generator: src/real/generate_innovation_dossiers.py